PHONETIC REDUCTION EFFECTS IN MALAYALAM

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INTRODUCTION TO THE STUDY

- Observing phonetic reduction as a function of speech rate
- 2 variables:
  - ”Enunciative Vowel”
  - Singleton, unaspirated, intervocalic plosives
- 3 speech rates:
  - Slow
  - Normal
  - Fast

INTRODUCTION TO THE STUDY

- Objectives:
  - Hierarchical model of lenition in Malayalam
    - Different levels of phonetic reduction – ð vs ɾ̪ vs d̪
  - Propose an updated rule for enunciative vowel distribution

INTRODUCTION TO THE STUDY

- Roadmap:
  - Malayalam Background
  - Literature Review
    - Enunciative Vowel
    - Consonant Lenition
  - Methods
  - Results and Discussion
    - Enunciative vowel analysis
    - Consonant lenition analysis

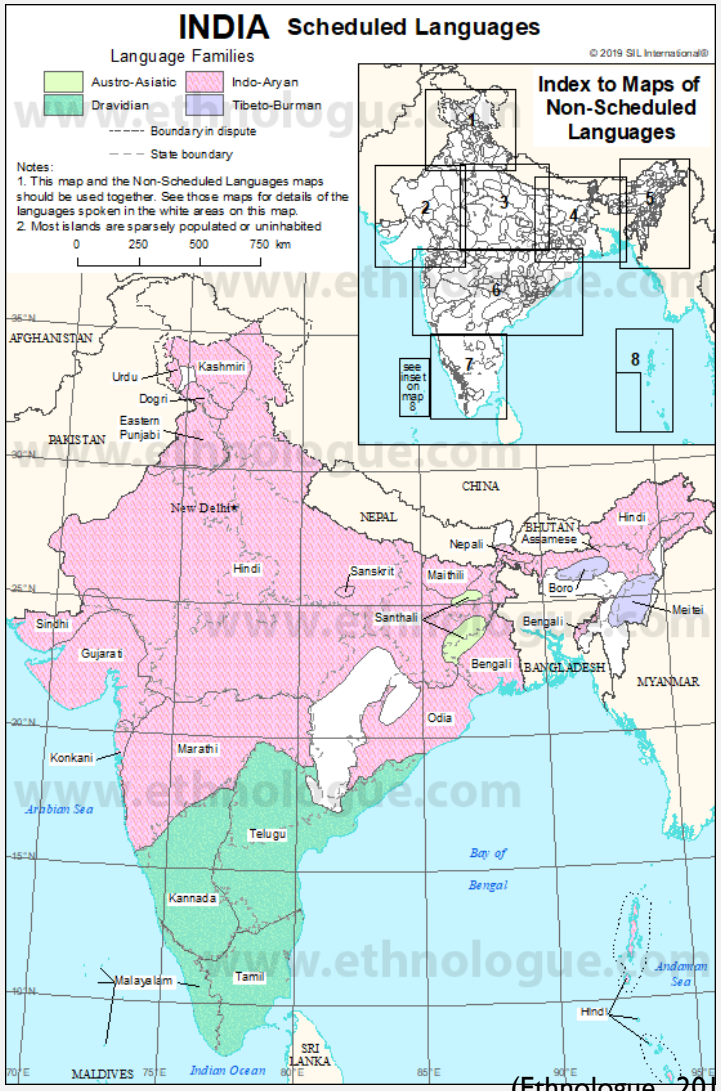
MALAYALAM

Dravidian language spoken in Kerala State.

Official language at the provincial level.

35.5 million speakers within India.

37.8 million speakers worldwide.

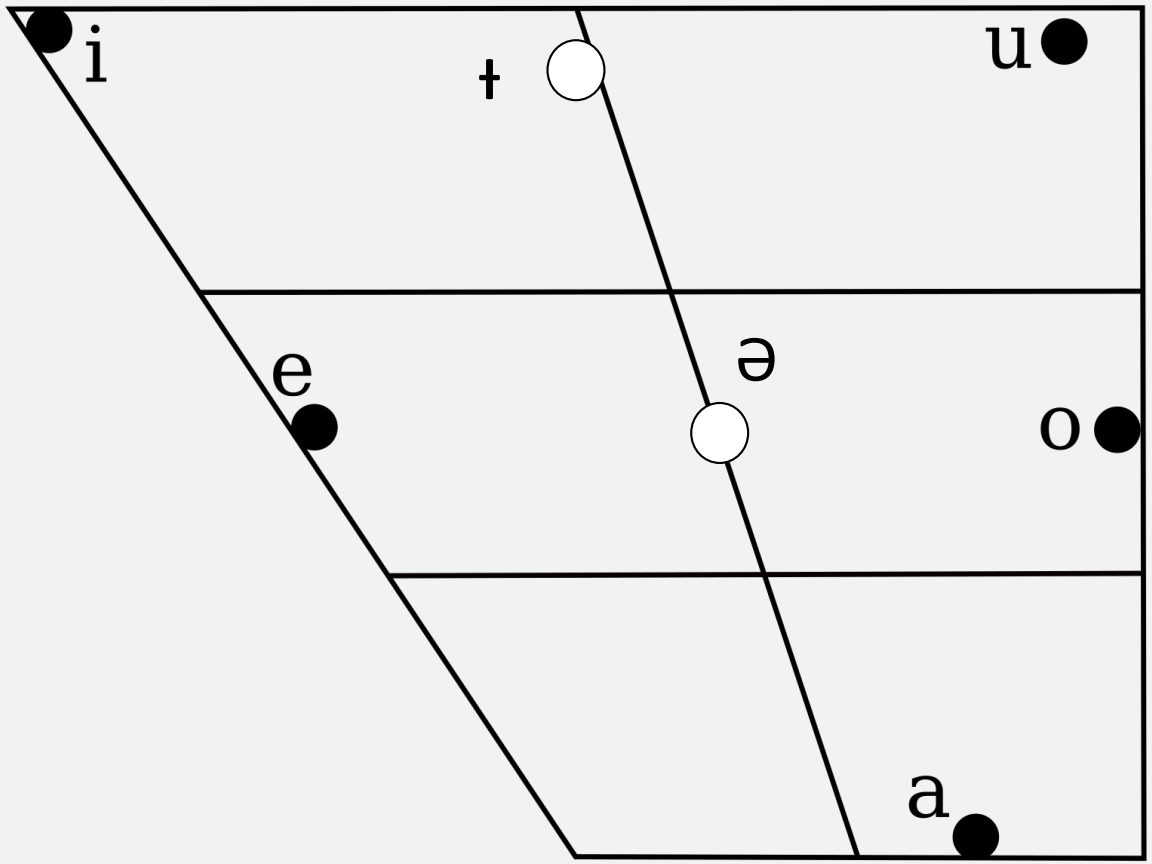


(Ethnologue, 2019)

MALAYALAM ENUNCIATIVE VOWEL

LITERATURE

- Maintains CV structure
  - Non-phonemic: word-medial, word-final
    - (Namboodiripad & Garellek, 2016)
  - Protects word final obstruents
  - Prevents C#C combinations
- “Schwa” or “Enunciative Vowel” insertion
  - (Mohanan, 1986; Valentine, 1976; Asher & Kumari, 1997)
- Reported as [ə], but can range in quality.
  - [ə] ~ [ɨ]
  - Always short
- Reportedly not present in C#V contexts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ɨ
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e
ə
o
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OPTIONALITY

- Enunciative vowel is perhaps *optional* in C# V context
  - [pod̪ap:ə on:um] ~ [pod̪ap: on:um]
- Short, unstressed vowels can be deleted (Asher & Kumari, 1997: 440)
  - Casual speech
  - Faster speech
- Enunciative vowel also found preceding pauses (Valentine, 1976)
  - We'll return to this idea, but for now:

QUESTION:

- Enunciative vowel is perhaps *optional* in C# V context
  - [pod̪ap:ə a:ɻ̩ka:ɳə] ~ [pod̪ap: a:ɻ̩ka:ɳə]
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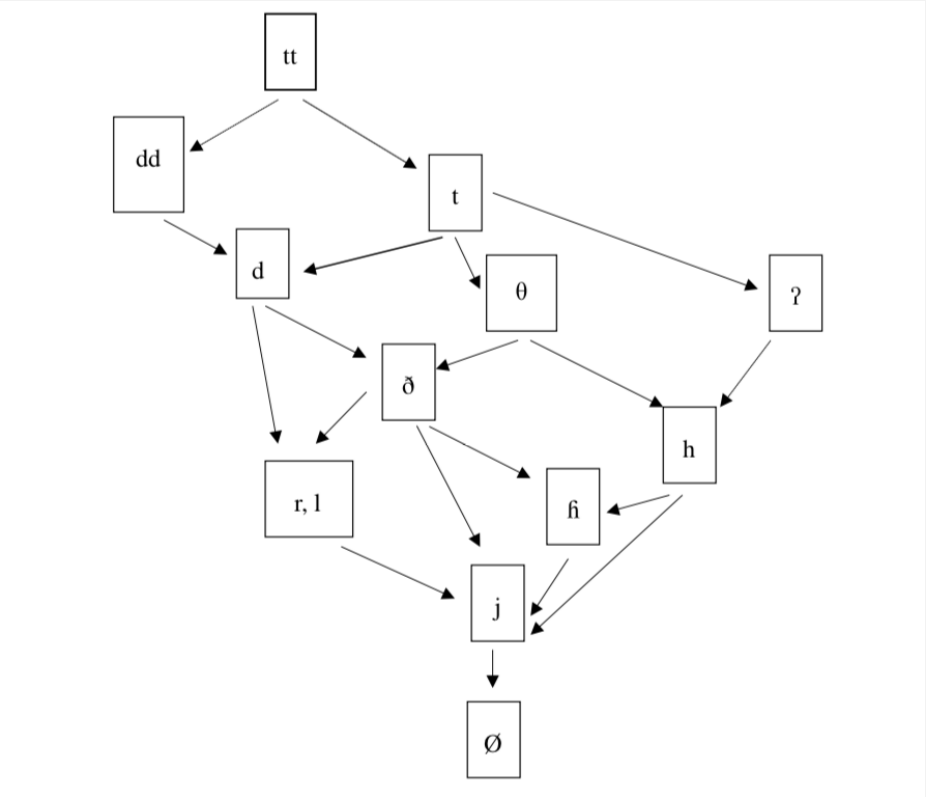
Does speech rate affect the presence of [ə] in optional contexts?

CONSONANT LENITION

LENITION: REDUCING ARTICULATORY EFFORT

Figure from Bauer (2008).



LENITION EXAMPLES (BAUER, 2008)

- Shortening of geminate consonants
  - Phonetically more “singleton-like”
  - Not the focus of current study
- Voicing of obstruents in intervocalic positions
  - Malayalam lacks voicing contrast in native words
    - Sanskrit borrowings maintain this distinction (Namboodiripad & Garellek (2016))
  - Orthography generally indicates voiceless stops intervocalically
  - For now, assume underlyingly voiceless

LENITION EXAMPLES (BAUER, 2008)

- Spirantization of obstruents in intervocalic positions
  - Change from plosive to fricative specifically
  - Plosive → fricative → approximant
- Debuccalization of stops word-finally (t → ʔ)
  - Lack of obstruent codas – Doesn't apply
- Segment deletion
  - Nasal deletion at boundaries (Mohanan,1986)
    - *maram* `tree' → marattinde `tree-GEN'
    - *malsyam* 'fish' → *malsyaccanta* 'fish market'
  - Obstruent deletion?

OTHER REPORTS -- MALAYALAM

- Voiced aspirated stops → voiceless aspirated stops
  - In casual speech (Namboodiripad & Garellek, 2016)
    - Lenition? Fortification?
- /k/ → /ɣ̞/, /h/, /ɦ/
  - Asher & Kumari (1997)
- Short, unstressed vowels can be omitted
  - Asher & Kumari (1997)

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Does lenition occur at different levels as a function of speech rate?

Does speech rate affect the presence of [ə] in optional contexts?

METHODS

METHODS

- Task
  - Read ”The North Wind and Sun” passage
    - From Namboodiripad & Garellek (2016)
- Speakers
  - 3 speakers analyzed for enunciative vowel use
    - 2 speakers read at all 3 rates
    - 1 speaker read at an unspecified rate (recording from JIPA)
      - (Namboodiripad & Garellek, 2016)
      - categorized as “normal”
  - 1 speaker analyzed for consonant lenition

METHODS

- Rate (subjective)
  - Slow
  - Normal
  - Fast
- Estimate of speech rate
  - Measured overall time with pauses omitted
    - Pauses between sentences, word boundaries between sonorants

PREDICTIONS

Enunciative Vowels

1. Fewer enunciative vowels as a function of rate
2. Enunciative vowel will be present in less frequently occurring words

Stop Lenition

1. Intervocalic stops will become more lenited at higher speech rates
    - More lenition with each increase in speed
    - Development of a hierarchical model through level separation
2. Place will have an effect on frequency and degree of lenition.

RESULTS & DISCUSSION

SPEECH RATES

	Slow	Normal	Fast
Speaker One	45 s	35 s	27 s
Speaker Two	46 s	36 s	28 s
Average	45.5 s	35.5 s	27.5 s
# Pauses*	23	18	8

* Measured only in Speaker One

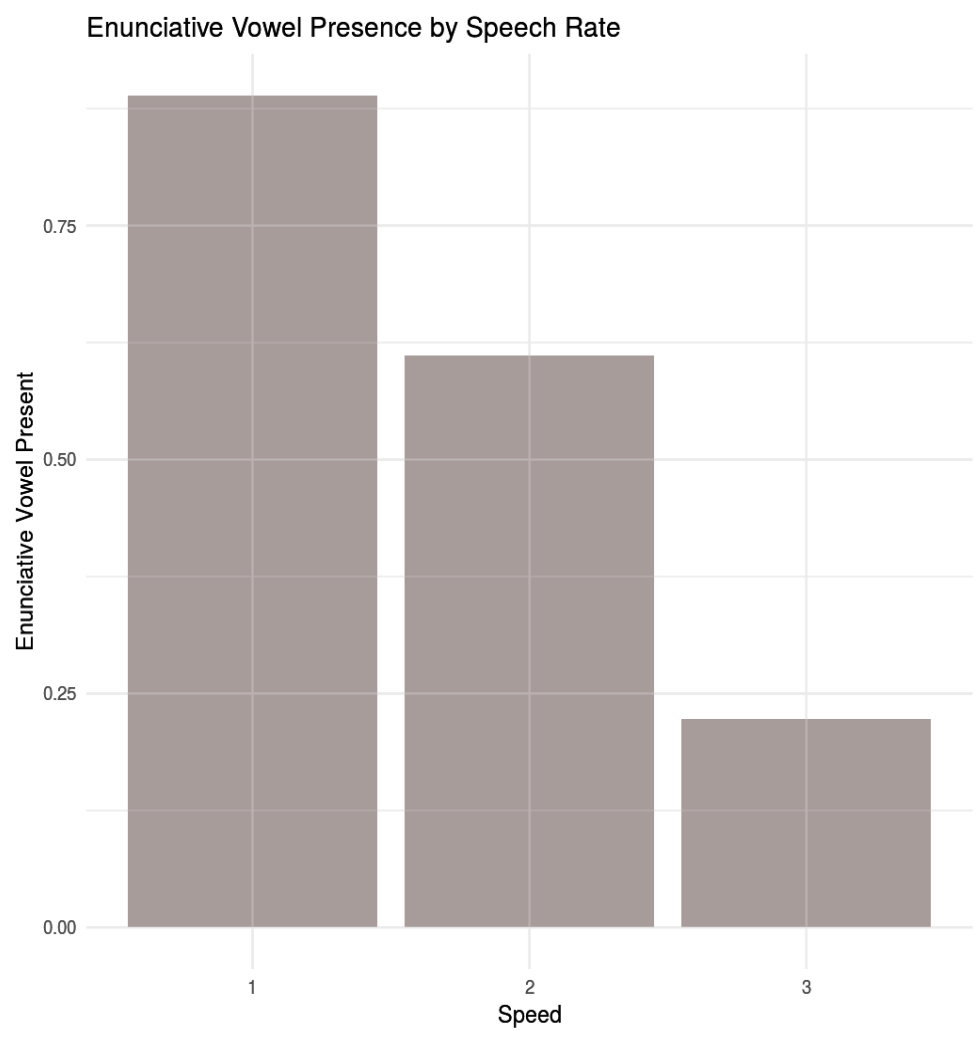
Does speech rate affect the presence of [ə] in optional contexts?

EU-DELETION:
C_V CONTEXT

Enunciative vowel [ə ~ ɨ] largely omitted in ”quick” speech.

Eu is deleted more frequently in unprompted speech than “slow” or “medium.”

Speed x Vowel Presence: p = < 0.001



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Enunciative Vowels

1. Fewer enunciative vowels as a function of rate ✓
2. Enunciative vowel will be present in less frequently occurring words ?

   Not enough tokens to determine any correlation

POSSIBLE EXPLANATIONS

1. Reduction occurs to the extent allowable by contrast in fast speech, and [ə] is non-contrastive.

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$$\emptyset \rightarrow [ə] \ / \ ___ \ \text{pause}$$ (Valentine 1976)

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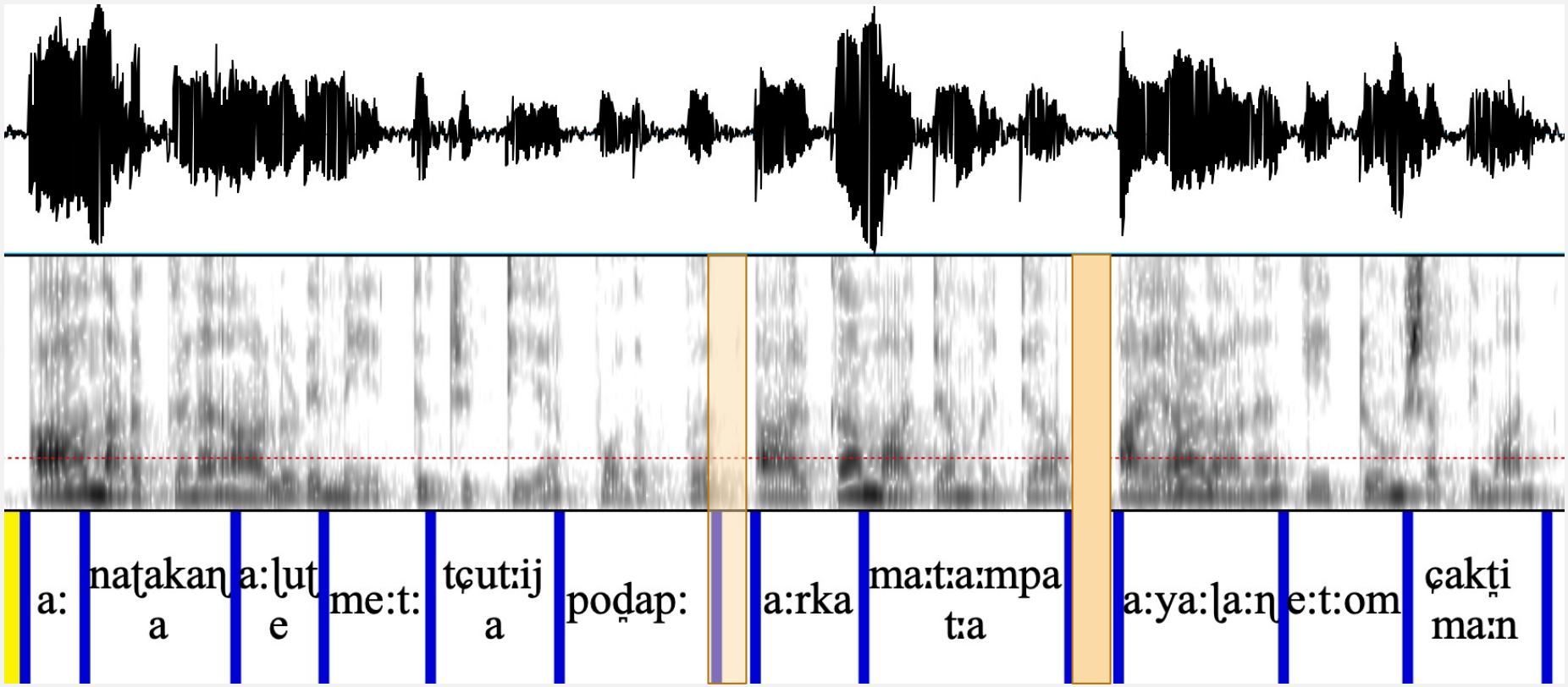
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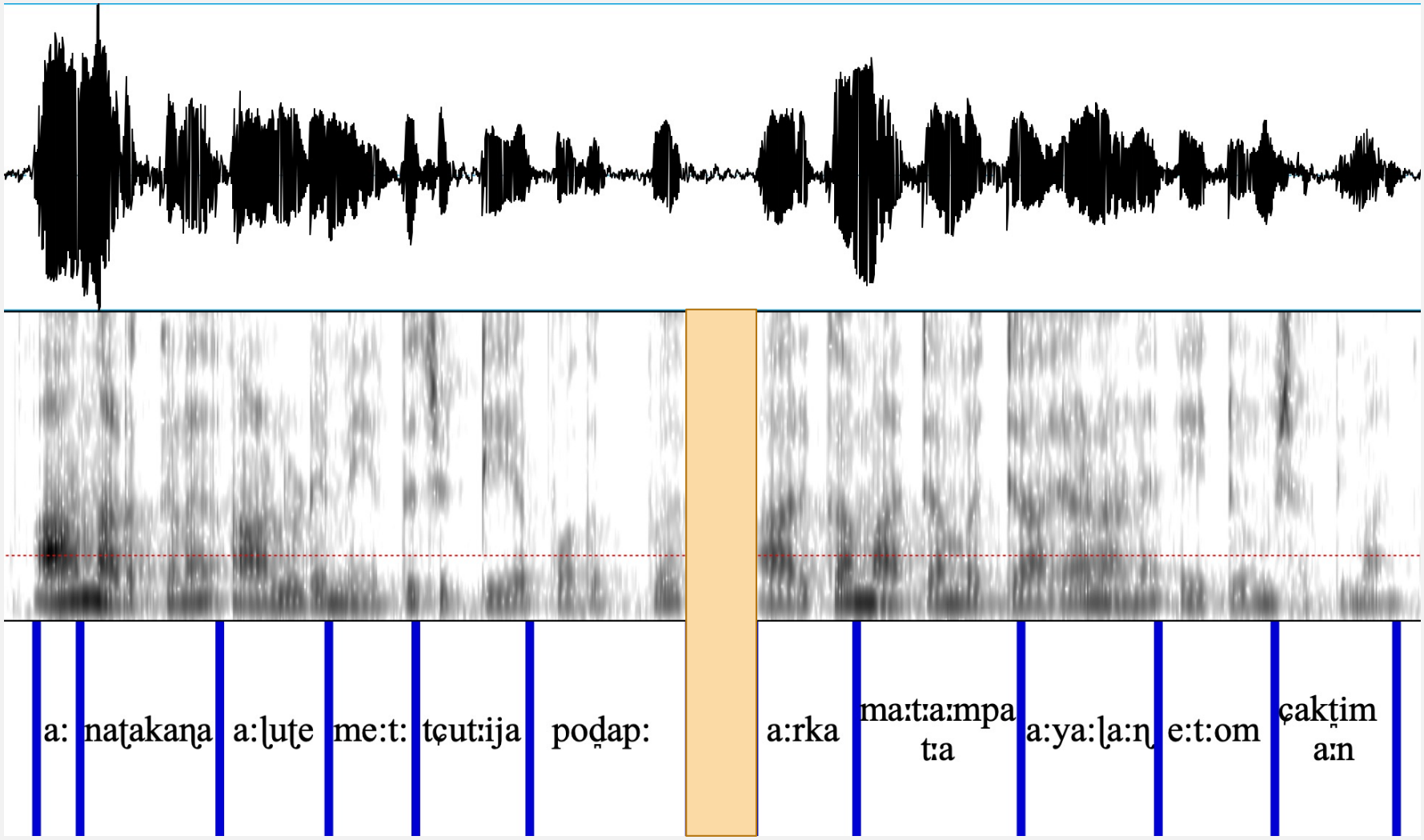
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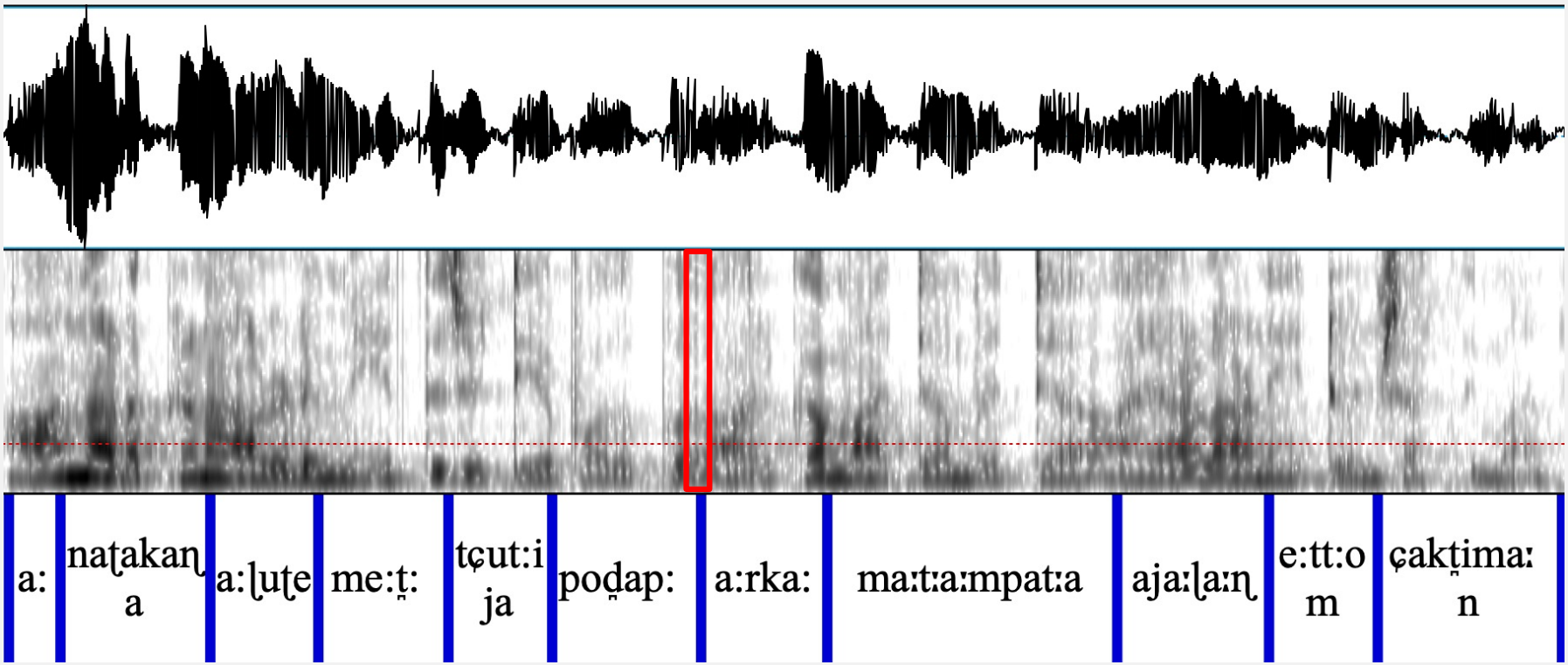
1. Reduction occurs to the extent allowable by contrast in fast speech, and [ə] is non-contrastive.

2. Higher speech rate means fewer pauses in speech.

   Ø → [ə] / ___ pause (Valentine 1976)

However, “pause” may not be enough to fully capture what’s happening here.

3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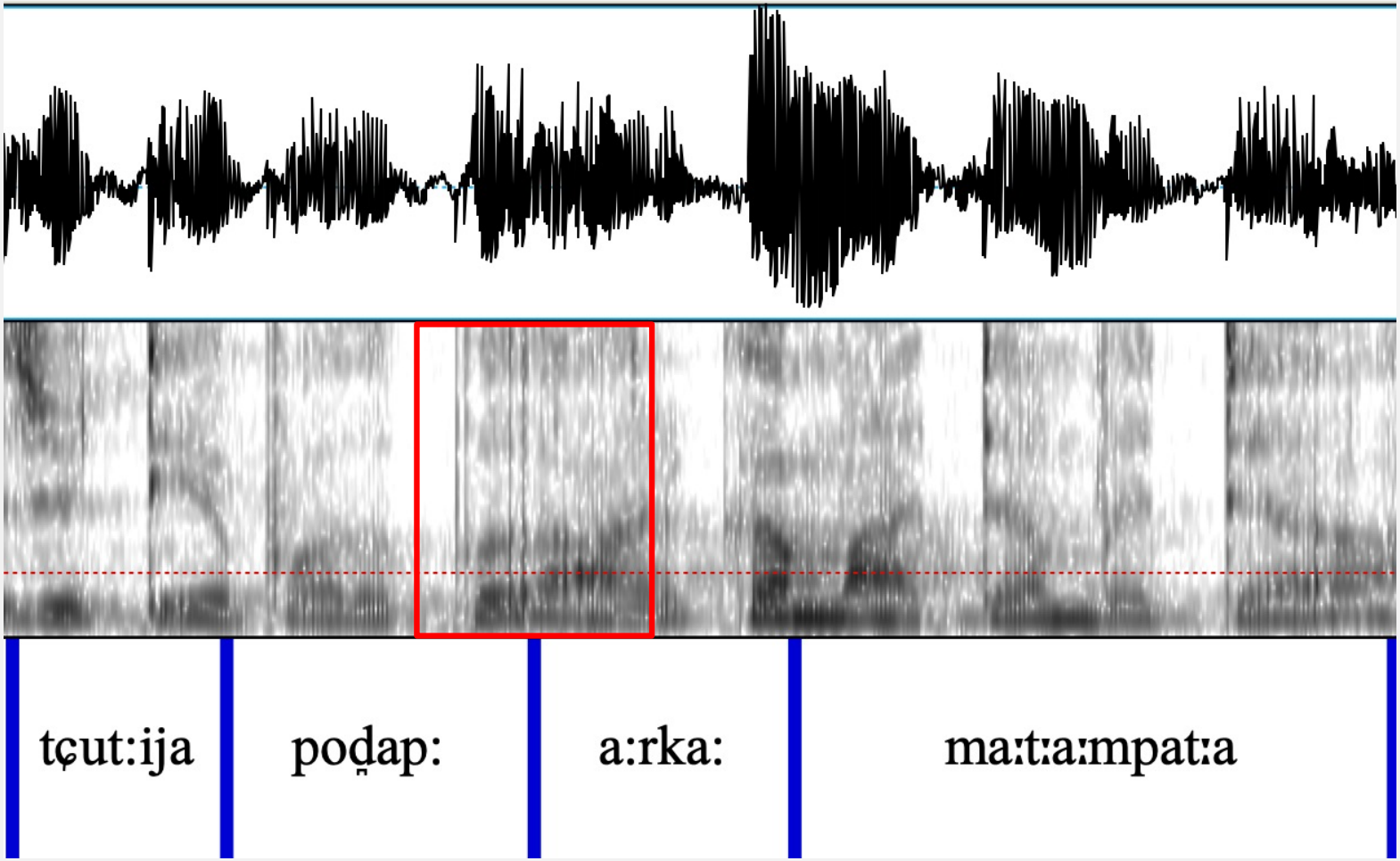

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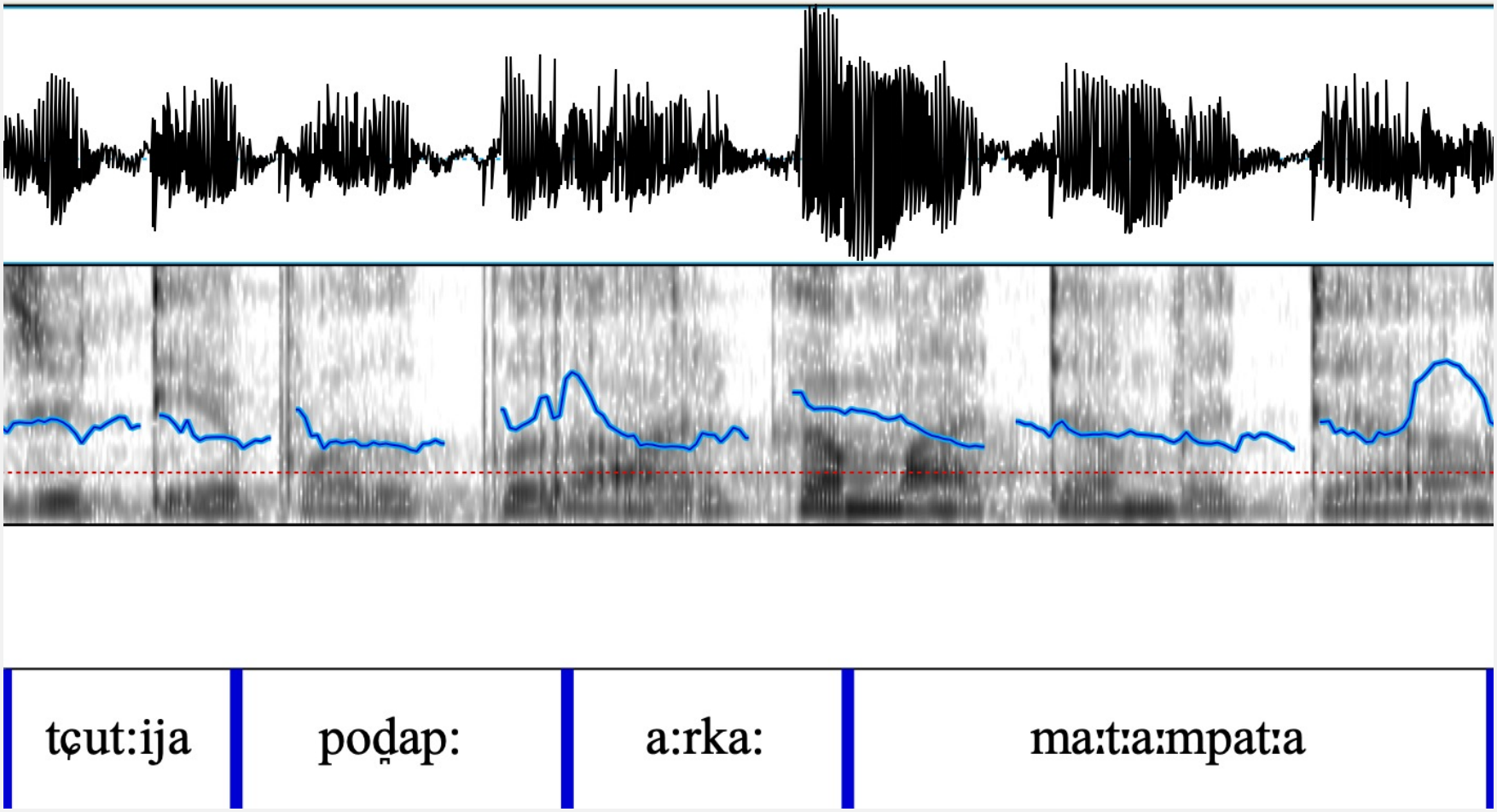

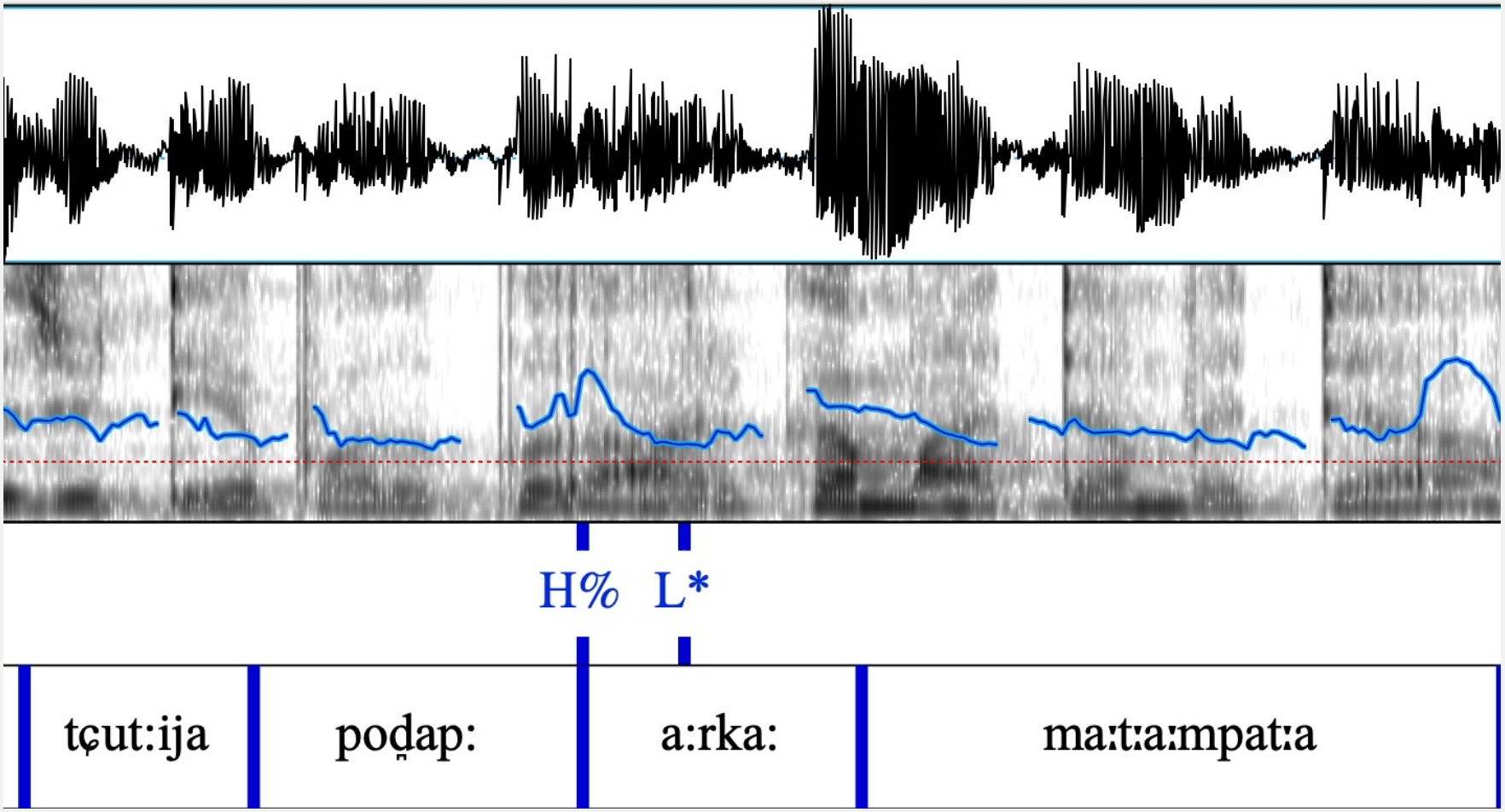
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L*
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POSSIBLE EXPLANATIONS

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2. Higher speech rate means fewer pauses in speech.

2a. Higher speech rate means fewer prosodic breaks.

Ø → [ə] / ___ pause (Valentine 1976)

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2a. Higher speech rate means fewer prosodic breaks.

Ø → [ə] / ___ %

LENITION

Approaching a hierarchical model

METHODS

- Tokens were coded visually according to a set of rules:
  - Stops are largely defined by preceding silence
  - Taps are defined by single burst
  - Fricatives require noise in upper formants
  - Approximants change formants between vowels
  - If none of these occur and the consonant is not heard, the consonant is considered deleted
  - Ambiguous tokens were not considered.

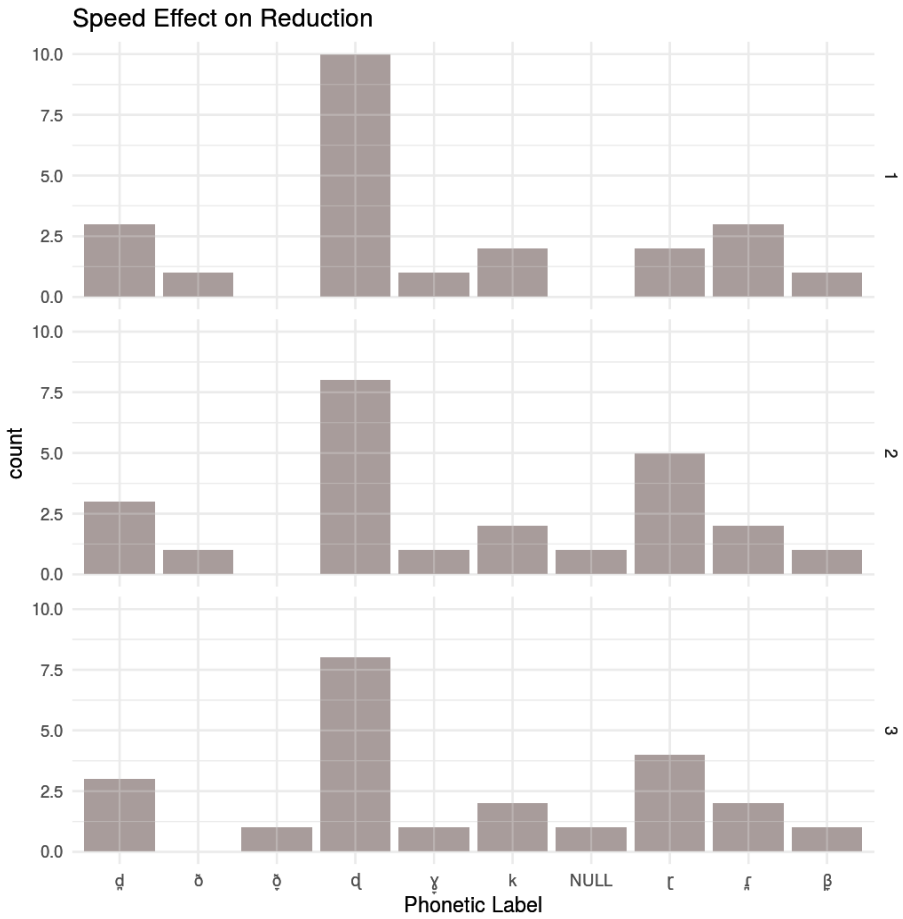
Does lenition occur at different levels as a function of speech rate?

STOP LENITION

Speed appears to have no effect on lenition levels.

No statistical analyses have been run.

Speed ignored from this point forward.



TOWARDS A LENITION MODEL

(p)	β̞				
(t̪)	d̪	ɾ̪	ð	ð̞	NULL
(ʈ)	ɖ	ɽ			
k	ɣ̞				

STOP LENITION

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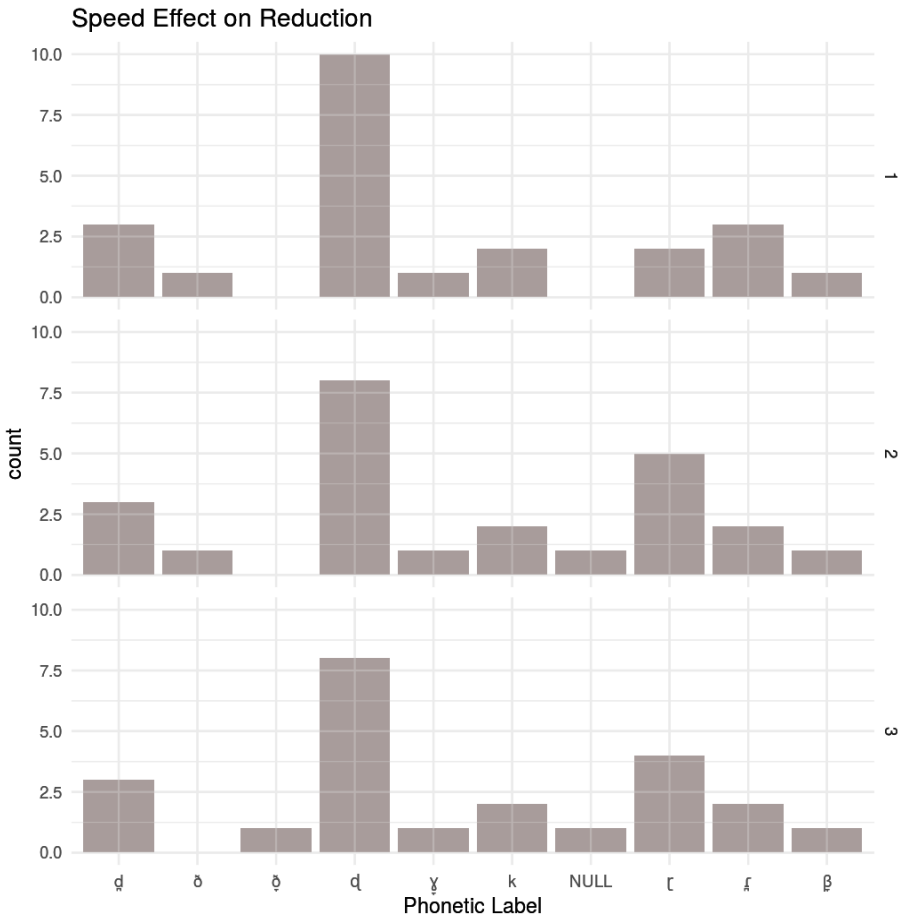
This assumes all consonants are equal.



PREDICTIONS

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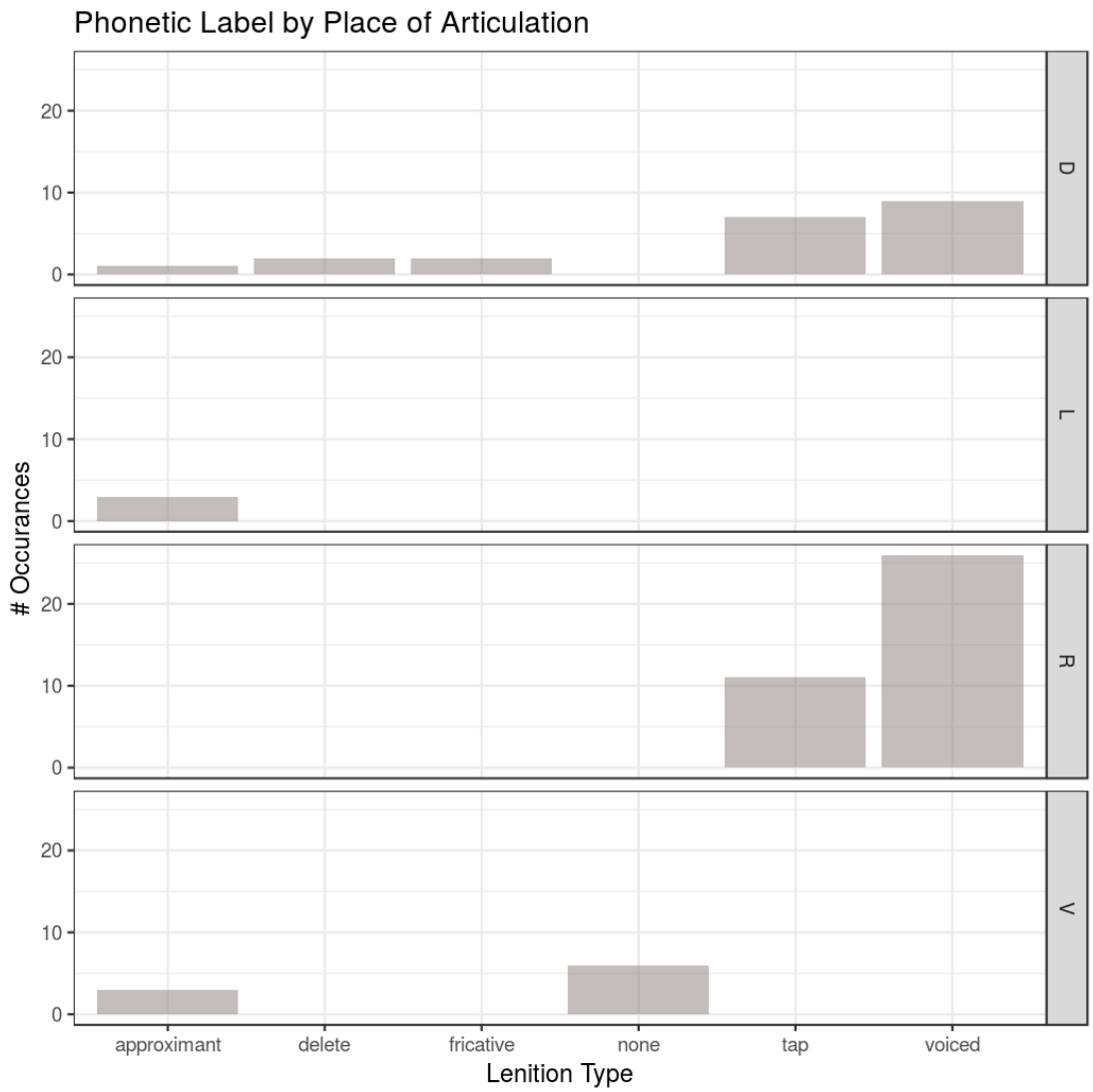
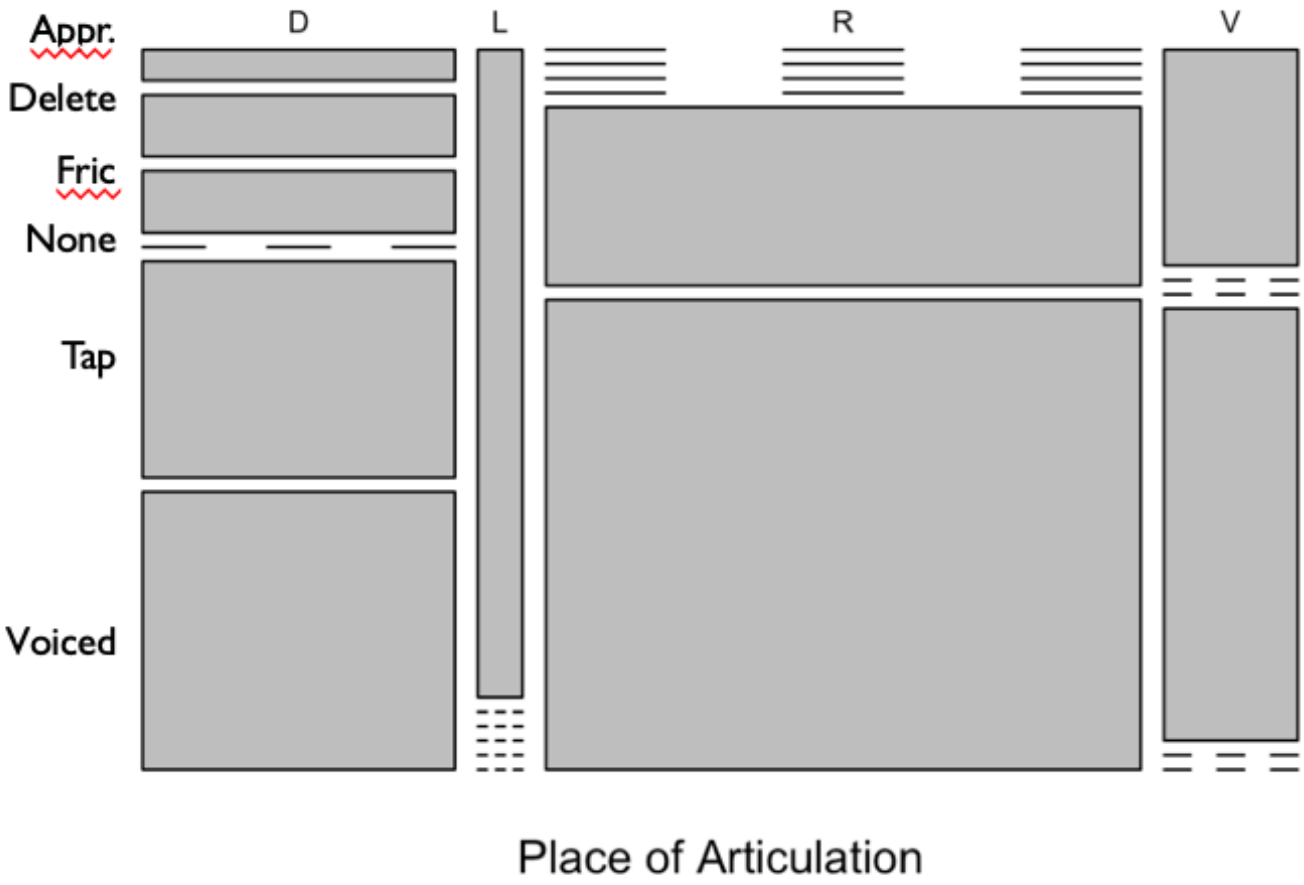

Phonetic Label by Place of Articulation
Occurances
Lenition Type
D
L
R
V
0
10
20
approximant
delete
fricative
none
tap
voiced

Place of Articulation effect on Lentition Type



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   - Again, too little data to tell.
   - Few velar and labial tokens.
   - Currently assumes all consonants are equal within word.

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   - Few velar and labial tokens.
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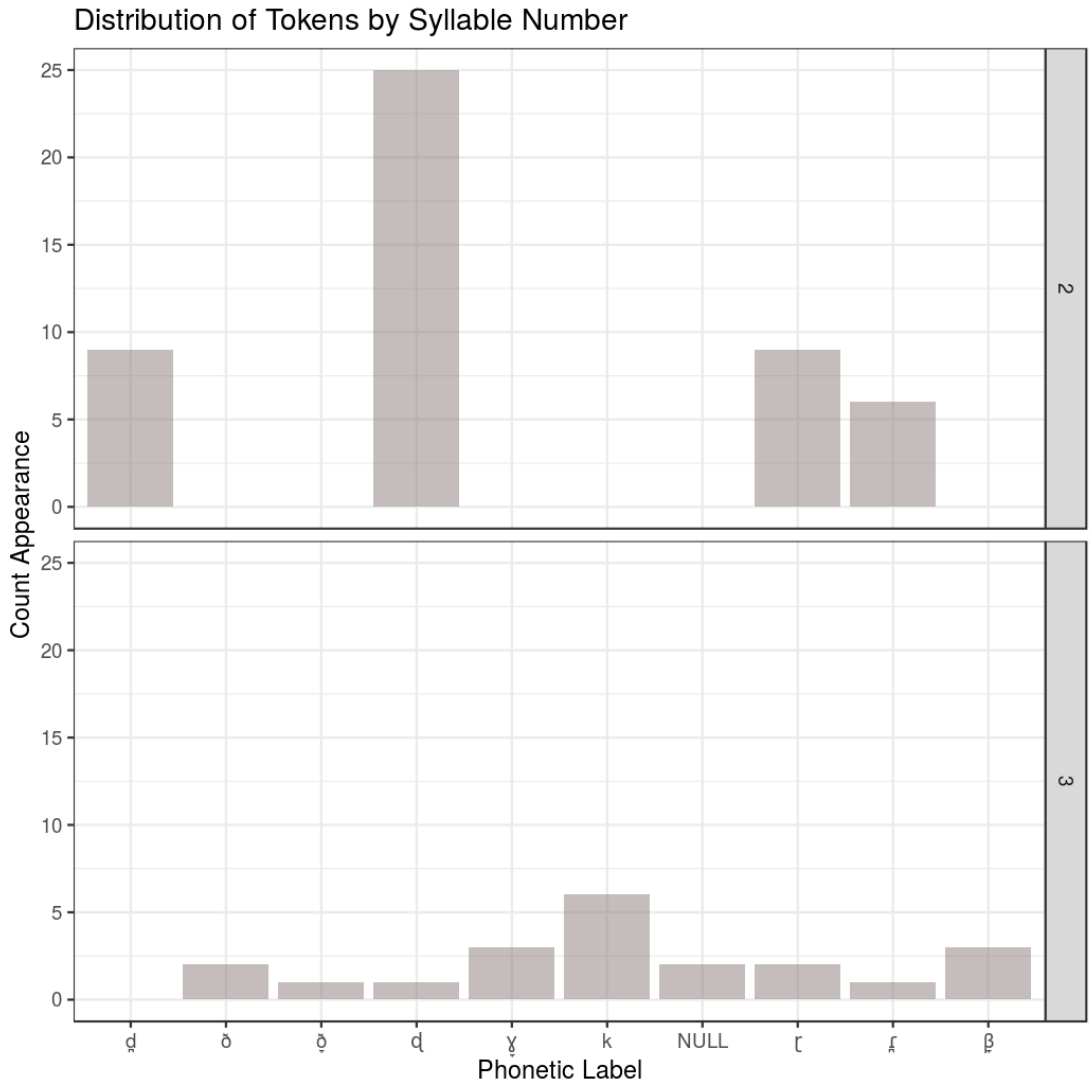
In the second syllable…

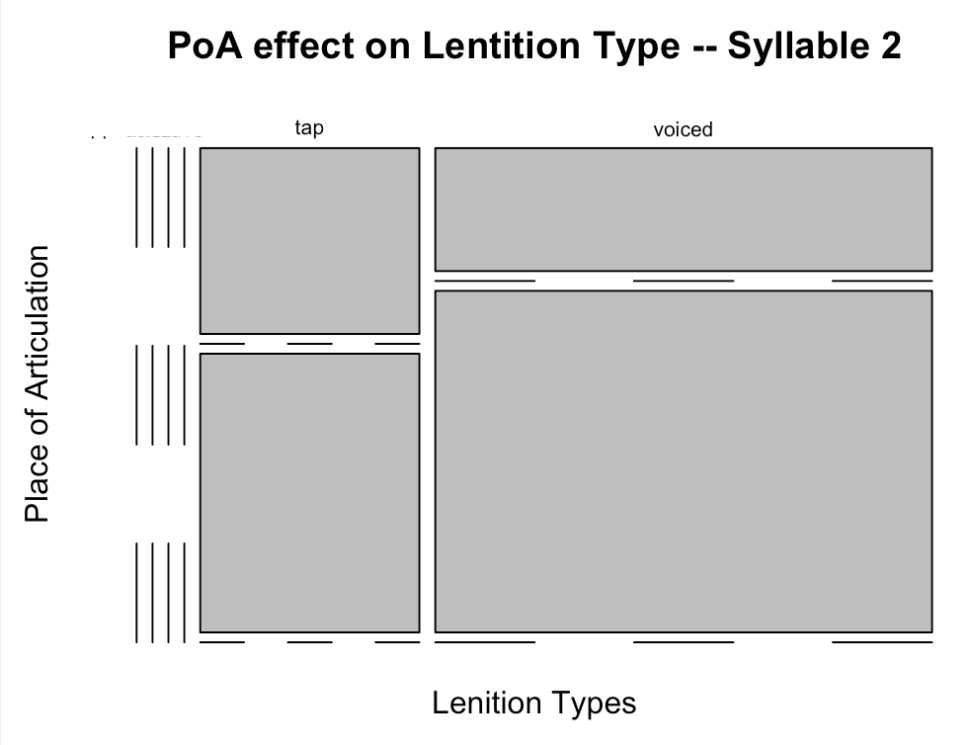
Dental & retroflex stops and taps occur after short vowels.

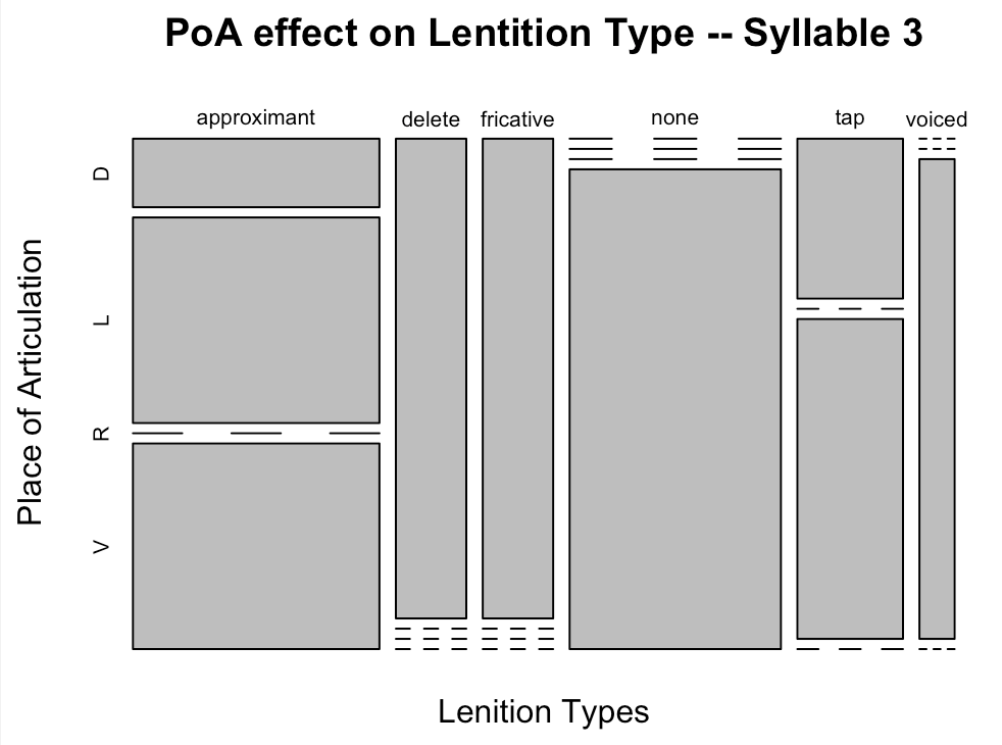
In the third syllable…
Wider distribution of lenited forms allowed.



STOP ALLOPHONES BY SYLLABLE

1st Syllable (initial)	2nd Syllable	3rd Syllable
p	N/A	β̞
t̪	d̪ / ɾ̪	ɾ̪ / ð / ð̞ / NULL
t	ɖ / ɽ	ɖ / ɽ
k	N/A	k / ɣ̞


PoA effect on Lentition Type -- Syllable 2
tap
voiced
Place of Articulation
Lenition Types


PoA effect on Lentition Type -- Syllable 3
approximant
delete
fricative
none
tap
voiced
D
L
R
V
Place of Articulation
Lenition Types

LENITION MODEL

Voiceless	1	2	3	4	5
Voiceless Stop	Voiced Stop	Tap	Fricative	Approximant	Deleted
(p)	b*			β̞	
(t̪)	d̪	ɾ̪	ð	ð̞	NULL
(ʈ)	ɖ	ɽ		ɻ̞*	

Parenthesis indicate "underlying form"
* Indicates form observed in data not reported here

LENITION MODEL

Voiceless	1	2	3
(p)	b*		β̞
(t̪)	d̪	ɾ̪	ð / ð̞ / NULL
(t)	ɖ	ɽ	ɻ*

Parenthesis indicate "underlying form"
* Indicates form observed in data not reported here

Thank you to the Malayalam speakers who participated in this experiment, Sameer ud Dowla Khan for advising this project, and all of you in attendance today!

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FUTURE DIRECTIONS

- Does one model better inform lenited speech?
  - Frequency of lenited speech by place in syllable, consonant place, etc.
- Distribution in continuous speech
  - Velar singletons
- Phonemic tap (ɾ̪ˠ) vs. dental tap vs. retroflex tap
- Internal / External frequency
  - Measured, no analyses run yet

STRESS

The presence of lexical stress is debated

This analysis assumes Mohanan's (1986) rules:

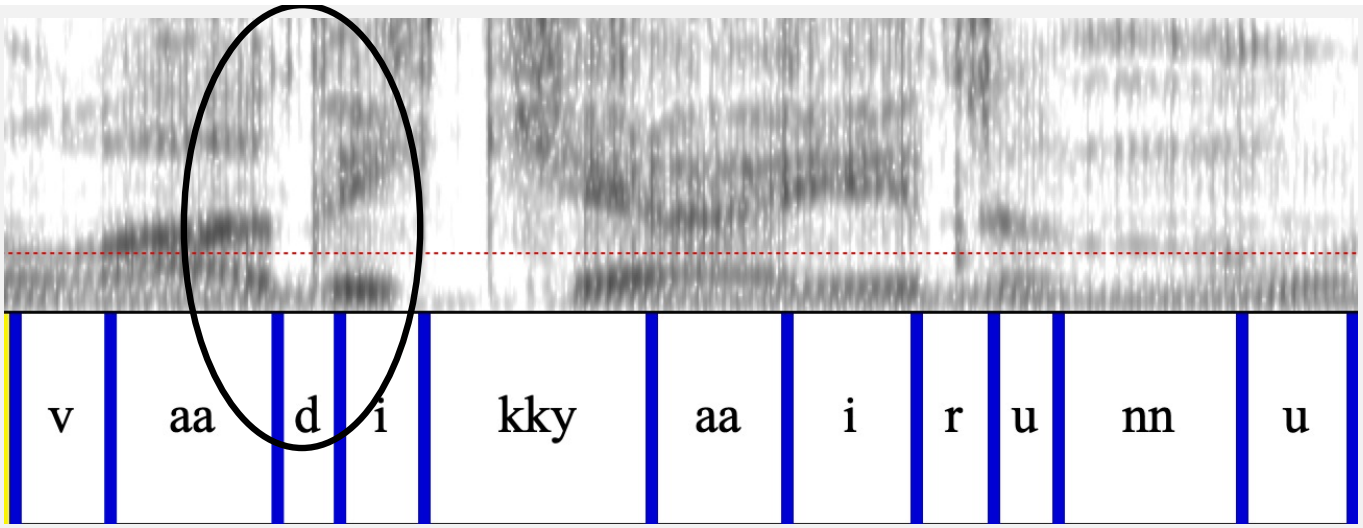
If a long vowel is in second syllable, that syllable holds stress.

If a short vowel is in the second syllable, first syllable holds stress.

The following uses length as an indicator of stress.

STOP ALLOPHONES AFTER LONG VOWELS

1st Syllable (initial)	2nd Syllable (V:___)	3rd Syllable (V:___)
p	N/A	N/A
t̪	d̪	N/A
t	ɖ / ɾ	N/A
k	N/A	N/A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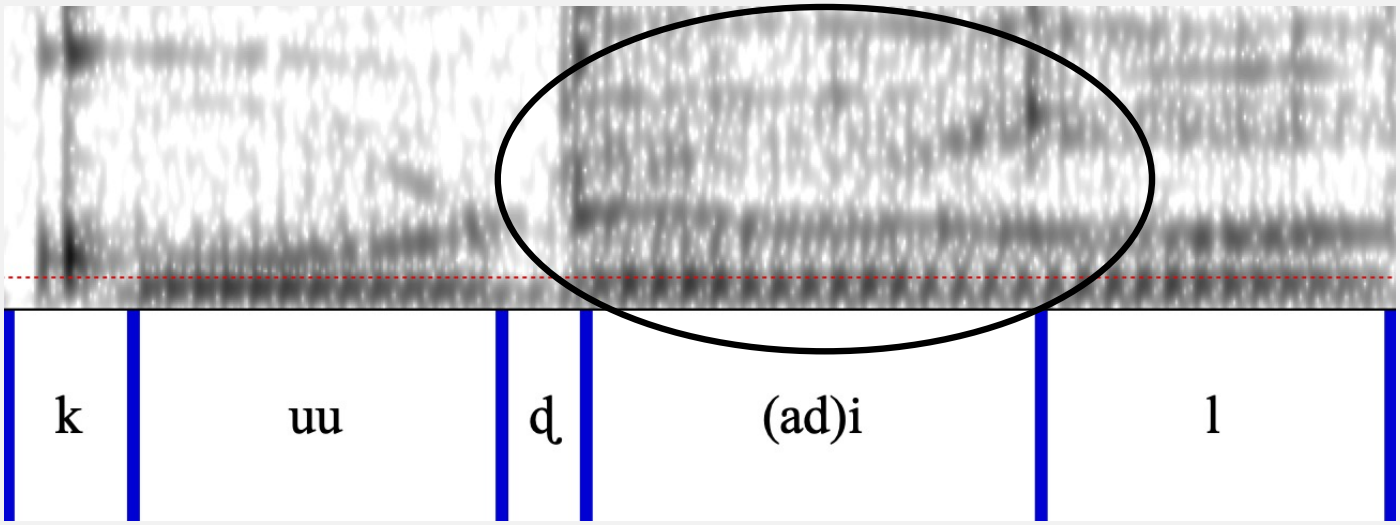
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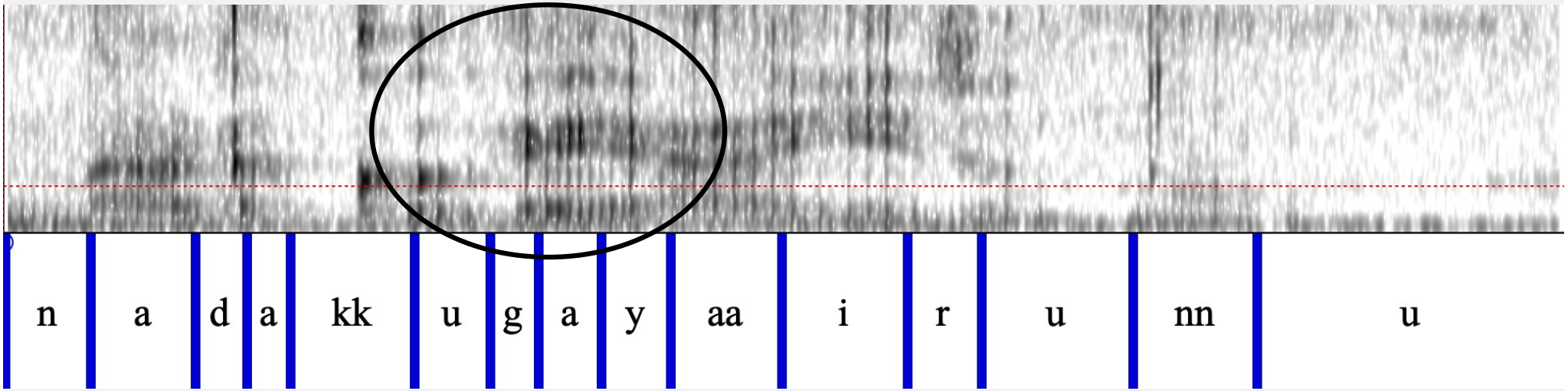
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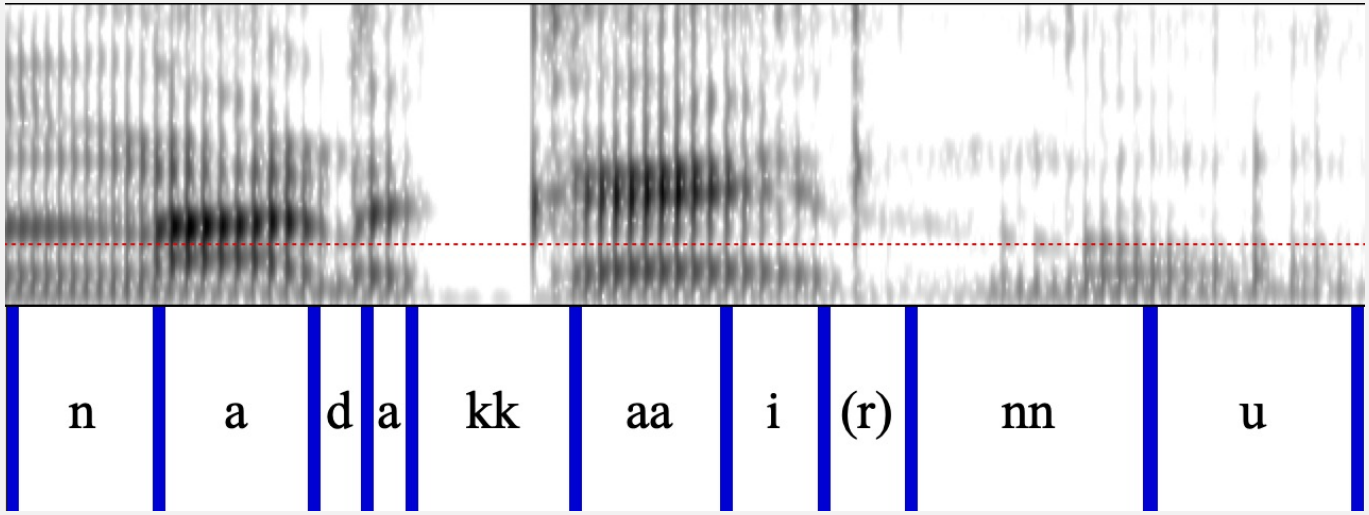
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